

Name



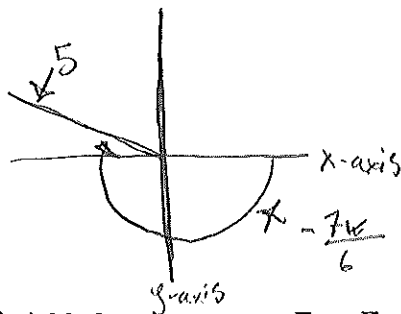
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Math 104 Spring 2025 Final Exam

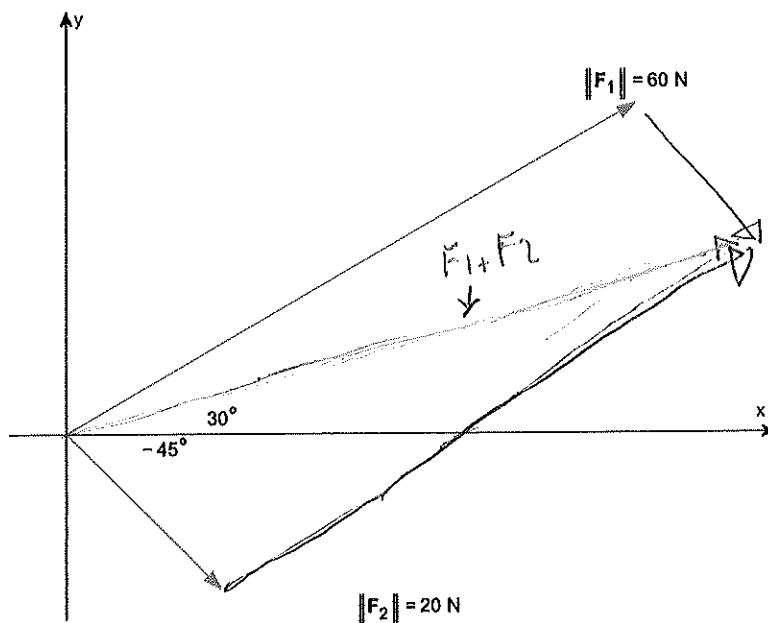
- A non graphing calculator is allowed
- Must show all work to receive credit
- Written or printed notes allowed (notes on digital devices not allowed)
- Attach all scratch paper

Question		Points
1	Polar, Vectors	/4
2	Average Rate of Change	/4
3	Trig Equations	/6
4	Side Side Angle Triangles	/6
5	Graphing, Domain, Range	/6
6	Applications	/6
Total		/30

1a) Plot the point is given in polar coordinates $(r, \theta) = \left(5, \frac{-7\pi}{6}\right)$



1b) Add the two vectors $F_1 + F_2$



$$F_1 = 30^\circ = \frac{\pi}{6} \rightarrow \begin{array}{c} 1 \\ \hline \sqrt{3} \\ \hline 2 \end{array} \cdot \frac{1}{2} = \frac{1}{2} \parallel 60 \parallel$$

$$F_2 = -45^\circ = -\frac{\pi}{4} = \begin{array}{c} \sqrt{2} \\ \hline 2 \end{array} \cdot \frac{\sqrt{2}}{2} = \frac{1}{2} \parallel 20 \parallel$$

$$F_1 + F_2 = (30\sqrt{3} + 30\sqrt{2})i + (30 - 30\sqrt{2})j$$

$$\rightarrow F_1 + F_2 = 30\sqrt{3} + 30\sqrt{2}i - 30\sqrt{2}j$$

$$F_1 = (1 \cdot 60) = 60$$

$$\rightarrow \frac{\sqrt{3}}{2} \cdot \frac{60}{1}$$

$$\rightarrow 30\sqrt{3}i$$

$$\rightarrow \frac{1}{2} \cdot \frac{60}{1} = 30$$

$$\rightarrow 30\sqrt{3}i + 30j$$

$$F_2 = (1 \cdot 20) = 20$$

$$\rightarrow \frac{\sqrt{2}}{2} \cdot \frac{60}{1} = 30\sqrt{2}i$$

$$\rightarrow \frac{\sqrt{2}}{2} \cdot \frac{60}{1} = -30\sqrt{2}j$$

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2) Let $g(x) = 4 - x^2$.

a) Find the average rate of change from -1 to 3 .

$$f_1 = g(-1) = 4 - (-1)^2 \quad m = \frac{f(b) - f(a)}{b - a}$$

$$\rightarrow g(-1) = 3$$

$$\rightarrow \frac{-5 - 3}{3 + 1}$$

$$f_2 = g(3) = 4 - (3)^2$$

$$\rightarrow g(3) = 4 - 9$$

$$\rightarrow \frac{-8}{4}$$

$$\rightarrow g(3) = -5$$

$$\rightarrow -2$$

b) Find an equation of the secant line containing $(-1, g(-1))$ and $(3, g(3))$.

$$y - y_1 = m(x - x_1)$$

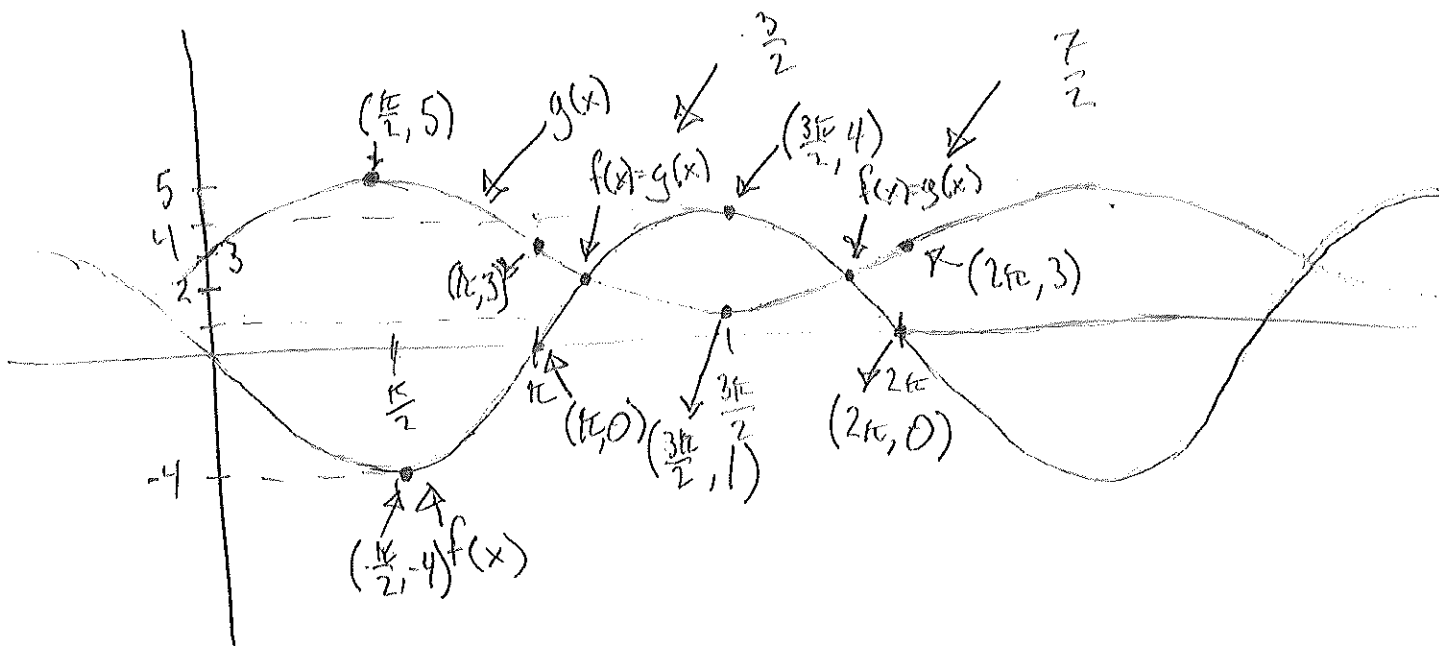
$$\rightarrow y - 3 = -2(x - (-1))$$

$$\rightarrow y - 3 = -2x - 2$$

$$\rightarrow y = -2x + 1$$

3) Let $f(x) = -4\sin(x)$ and let $g(x) = 2\sin(x) + 3$.

Graph $f(x)$ and $g(x)$ on the same axes.



Solve $f(x) = g(x)$ on the interval $[0, 2\pi]$.

$$f(x) = g(x)$$

$$\rightarrow -4\sin(x) = 2\sin(x) + 3$$

$$\rightarrow -4\sin(x) - 3 = 2\sin(x)$$

$$\rightarrow \frac{-4\sin(x) - 3}{2} = \sin(x)$$

$$\rightarrow \sin^{-1}\left(\frac{-4\sin(x) - 3}{2}\right)$$

$$\rightarrow \frac{-4 - 3}{2}$$

$$\rightarrow \left| \frac{-7}{2} \right|$$

$$\rightarrow \frac{7}{2}, \frac{3}{2}$$

$$\rightarrow -4\sin(x) = 2\sin(x) + 3$$

$$\rightarrow \frac{2\sin(x) + 3}{-4} = \sin(x)$$

$$\rightarrow x = \sin^{-1}\left(\frac{2\sin(x) + 3}{-4}\right)$$

$$\rightarrow x = \frac{2 + 3}{-4}$$

$$\rightarrow \frac{6}{-4}$$

$$\rightarrow \left| -\frac{3}{2} \right|$$

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4) Two sides and an angle are given. Determine whether the given information results in one triangle, two triangles, or no triangle at all. Solve any resulting triangle(s) and draw the triangle(s).

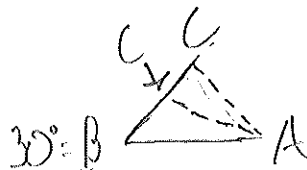
$$b = 4, c = 5, B = 30^\circ$$

$$\frac{\sin(30^\circ)}{4} = \frac{\sin(C)}{5}$$

$$180 - 38.68 = 141.32 + 30 = 171.32^\circ \checkmark$$

$$\rightarrow 5 \sin(30^\circ) = 4 \sin(C)$$

$$\rightarrow \frac{5 \sin(30^\circ)}{4} = \sin(C)$$



$$\rightarrow \sin^{-1}\left(\frac{5 \sin(30^\circ)}{4}\right) = C$$

$$\rightarrow C_1 \approx 38.6821 \text{ or } \approx 38.68^\circ$$

$$\rightarrow C_2 \approx 141.32^\circ$$

$$\rightarrow A_1 = 180 - (30 + 38.68)$$

$$\rightarrow A_1 = 180 - 68.68$$

$$\rightarrow A_1 = 111.32^\circ$$

$$\rightarrow A_2 = 180 - (141.32 + 30)$$

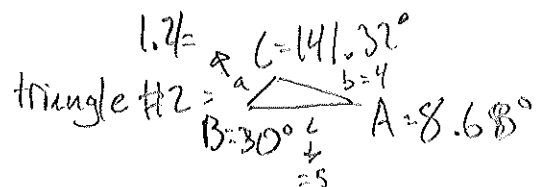
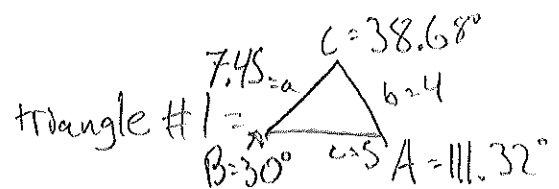
$$\rightarrow A_2 = 180 - 171.32$$

$$\rightarrow A_2 = 8.68^\circ$$

$$a_1 = \frac{\sin(30^\circ)}{4} = \frac{\sin(111.32^\circ)}{a}$$

$$\rightarrow a_1 = \frac{4 \sin(111.32^\circ)}{\sin(30^\circ)}$$

$$\rightarrow a_1 \approx 7.45$$

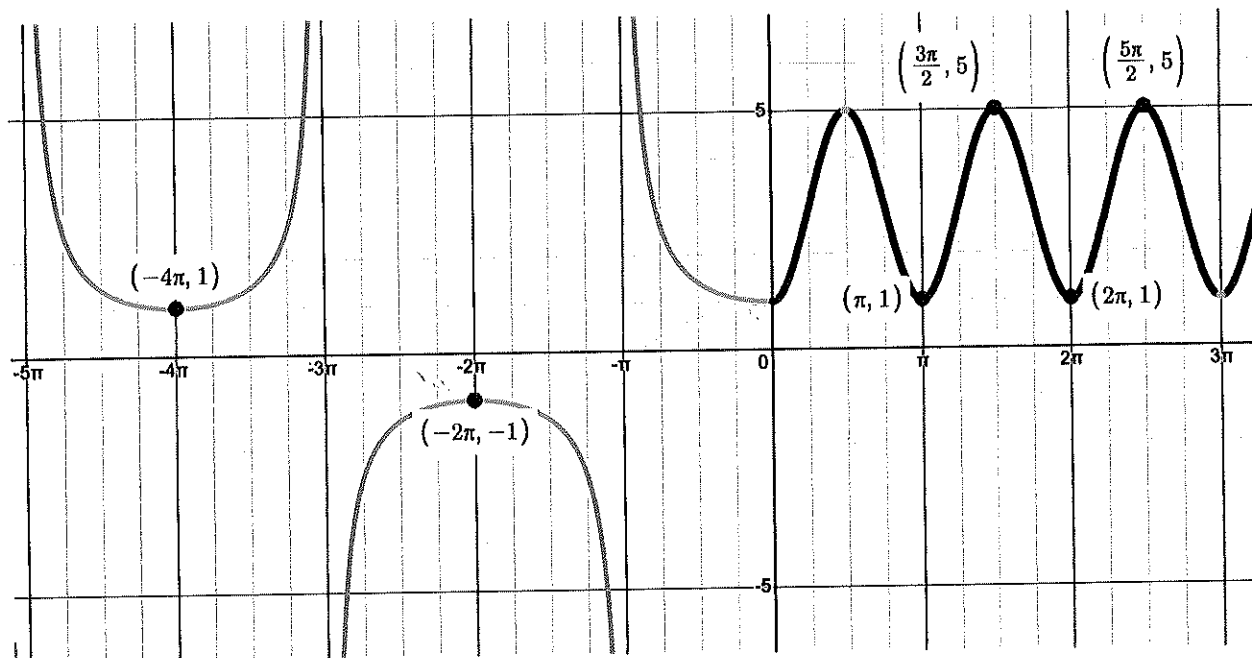


$$a_2 = \frac{\sin(30^\circ)}{4} = \frac{\sin(8.68^\circ)}{a}$$

$$\rightarrow a_2 = \frac{4 \sin(8.68^\circ)}{\sin(30^\circ)}$$

$$\rightarrow a_2 \approx 1.2$$

5) Given the graph below find the function $f(x)$ and state the domain and range.



$$f(x) = \begin{cases} \text{when } x < 0 \rightarrow \sec\left(\frac{1}{2}x\right) \\ \text{when } x \geq 0 \rightarrow -2\cos(2x) + 3 \end{cases}$$

Range = $\mathbb{R}$

Domain = $(-\infty, 0)$ except any odd integer $\cdot \pi$
 $\cup [0, \infty)$

black $f(x) =$

$$\rightarrow a = \frac{1}{2}(5-1) \quad b = \frac{2\pi}{\pi}$$

$$\rightarrow \frac{4}{2}$$

$$\rightarrow 2$$

$$\rightarrow 2 = a$$

$$\rightarrow k = \frac{1}{2}(5+1)$$

$$\rightarrow \frac{6}{2}$$

$$\rightarrow k = 3$$

red $f(x) =$

$$\rightarrow a = \frac{1}{2}(1+1)$$

$$\rightarrow a = 1$$

$$k = \frac{1}{2}(1+(-1))$$

$$\rightarrow k = \frac{0}{2} = 0$$

$$b = \frac{2\pi}{4\pi} = \frac{1}{2}$$

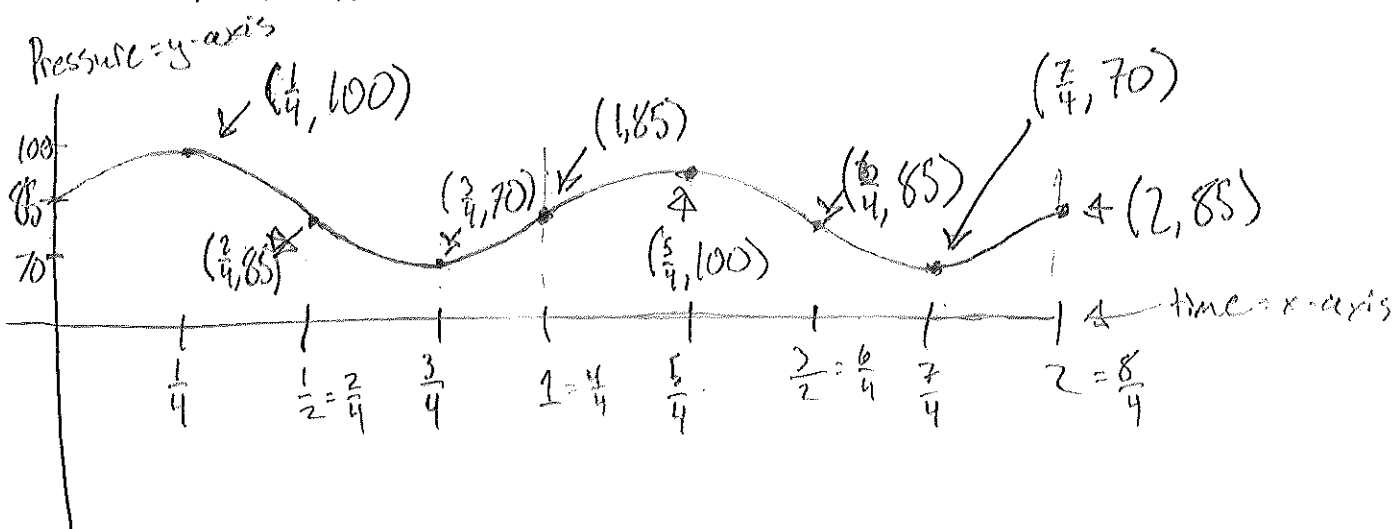
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6) Blood pressure is modeled by the function $P(t) = 15 \sin(2\pi t) + 85$ where the maximum value of P is the systolic pressure (which is the pressure when the heart beats), the minimum value is the diastolic pressure, and t is time in seconds.

$$\text{Period} = \frac{2\pi}{2\pi} = 1$$

a) Graph $P(t)$ for two periods, label your axes and local maximums.



heart rate: 1 sec for 1 beat (period = 1)

→ 60 sec in 1 min

→ $60 \div 1 = 60$

b) The systolic pressure is 100

c) The diastolic pressure is 70

d) The heart rate is 60 beats per minute